# When Does Temporal Memory Help? Selective SSM vs. MLP in RL-Based Multi-Target DOA Track Management

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***Abstract*—We address when temporal memory helps in reinforcement learning (RL)-based multi-target DOA track management. We prove that a selective SSM encoder outperforms a memoryless MLP if and only if the observation is *not* a sufficient statistic for the policy-relevant history—a condition satisfied by raw high-dimensional COP spectra over long episodes but not by hand-crafted low-dimensional features. We instantiate this as Mamba-COP-RL: a Mamba SSM encoder coupled with PPO, operating on the 181-dim COP spatial spectrum. On CombatTrackEnv (four combat scenarios), Mamba-COP-RL improves GOSPA by up to $25.2\%$ over fixed GM-PHD thresholds and $6.7\%$ over MLP-PPO, with gains ordered by temporal dependency horizon as predicted. The 41.4 KB INT8 model runs at $3.4$ ms/scan on STM32H7 Cortex-M7.**

***Index Terms*—direction of arrival, multi-target tracking, reinforcement learning, selective state space model, Mamba, GM-PHD, edge AI, CombatTrackEnv, sufficient statistic**

## I. Introduction

Multi-target DOA tracking with compact arrays is a core challenge in defense, speech enhancement, and wireless communications. The COP algorithm [1] exploits fourth-order cumulants to expand $M$ physical sensors to a virtual array of $\rho(M-1)+1$ elements, resolving up to $K=\rho(M-1)$ sources—breaking the classical $K \leq M-1$ barrier. Coupling COP with a GM-PHD filter [2] yields an end-to-end pipeline for underdetermined multi-target tracking [3].

The GM-PHD filter is governed by three scalar hyperparameters: birth weight $w_b$, pruning threshold $\tau_p$, and detection probability $p_D$. Fixed values degrade sharply in non-stationary combat environments: jamming collapses SNR, stealth targets vanish and reappear, saturation attacks flood the scene, and formations fragment. RL offers a data-driven solution—map filter state to parameters adaptively.

Prior RL-based filter tuning uses memoryless MLP policies [4], [5]. A natural extension is to add temporal memory (LSTM, GRU, SSM) to the policy. However, recurrent policies add complexity and can *hurt* performance if temporal information is already saturated by hand-crafted summary statistics.

**Central question**: *when does temporal memory help?*

**Our answer** (Proposition 1): SSM outperforms MLP when the observation is *high-dimensional raw signal* and the episode is *long* enough to create multi-scan causal dependencies. We verify this with a controlled ablation:

- **12-dim** hand-crafted statistics, 40 scans $\Rightarrow$ MLP wins; recurrent policies hurt.
- **183-dim** raw COP spectrum, 200 scans $\Rightarrow$ Mamba wins on Stealth and Jamming.

**Contributions**:

1) A proposition with complete proof characterizing when SSM temporal memory improves RL-based track management.
2) **Mamba-COP-RL**: selective SSM encoder + PPO actor-critic; 42 K params, 41.4 KB INT8.
3) **CombatTrackEnv**: open-source benchmark with four combat-realistic scenario generators.
4) Comprehensive ablation including GRU-PPO, LSTM-PPO, d_state sensitivity, and episode-length dependency.
5) Edge deployment analysis: $3.4$ ms/scan on STM32H7 Cortex-M7.

## II. Related Work

Prior RL-based GM-PHD tuning uses memoryless MLP policies with low-dimensional hand-crafted features [4], [5]; Ewers *et al.* [6] compared BiGRU and self-attention against MLP for radar tracking RL but did not formally characterize when recurrence helps. Hausknecht and Stone [7] showed DRQN improves performance in POMDPs; Dorka [8] analyzed generalization limits of recurrent PPO. Mamba [9] extends S4 [10] with input-dependent selectivity, enabling adaptive gating of irrelevant scans (e.g., zero-signal stealth windows) while retaining trajectory history in $\mathbf{h}_t$—a capability absent from fixed-rate GRU/LSTM. Ota [11] applied Mamba to offline RL; Han *et al.* [12] to audio signal processing. We are the first to deploy Mamba as an *online* RL encoder for adaptive DOA filter tuning and to formally characterize when temporal memory helps.

## III. Background

### A. COP-DOA and GM-PHD Tracking

The COP pseudo-spectrum is:

$$P_{\mathrm{COP}}(\theta) = \frac{1}{\|\mathbf{P}_{\mathbf{A}}^{\perp}\mathbf{a}_v(\theta)\|^2} \tag{1}$$

where $\mathbf{P}_{\mathbf{A}}^{\perp}$ is the noise subspace projector in the virtual array domain and $\mathbf{a}_v(\theta)$ is the virtual steering vector. Evaluated at $N_\theta=181$ angles, $P_{\mathrm{COP}}$ is the primary observation for our RL policy.

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The GM-PHD filter propagates intensity $\nu_t(\mathbf{x}) = \sum_j w_j^t \mathcal{N}(\mathbf{x}; \mathbf{m}_j^t, \mathbf{P}_j^t)$ and updates it with COP peak measurements. Three hyperparameters govern behavior: $w_b$ (birth weight), $\tau_p$ (prune threshold), and $p_D$ (detection probability).

### B. Selective State Space Models

A selective SSM [9] defines:

$$\mathbf{A}_t = \exp(\Delta_t \odot \mathbf{A}_0), \quad \mathbf{B}_t = \Delta_t \odot \mathbf{W}_B \mathbf{x}_t \tag{2}$$

$$\mathbf{h}_t = \mathbf{A}_t \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{x}_t, \quad y_t = \mathbf{c}_t^\top \mathbf{h}_t \tag{3}$$

where $\Delta_t, \mathbf{B}_t, \mathbf{c}_t$ are input-affine functions. Selectivity allows the model to learn *which* inputs to retain in $\mathbf{h}_t$ and which to discard, enabling long-range compression without fixed forgetting rates—a key advantage over GRU and LSTM.

## IV. Mamba-COP-RL

### A. When Does Temporal Memory Help?

**Proposition 1** (SSM Advantage Condition)**.** *Let $f : \mathbb{R}^{d_o} \to \mathbb{R}^{d_f}$ be any deterministic observation summary ($d_f \le d_o$). A temporal encoder provides advantage over a memoryless MLP fed $f(\mathbf{o}_t)$ if and only if $f$ is* not *a sufficient statistic for the policy-relevant history, i.e.,*

$$\mathcal{I}(\mathbf{o}_t;\, \mathbf{o}_{1:t-1}) \;>\; \mathcal{I}(f(\mathbf{o}_t);\, \mathbf{o}_{1:t-1}) \tag{4}$$

*where $\mathcal{I}(\cdot;\cdot)$ denotes mutual information.*

*Proof sketch.* (*Necessity.*) If equality holds, $f$ is a sufficient statistic by the data processing inequality [13], so the MLP on $f(\mathbf{o}_t)$ already accesses all history-relevant information and no temporal encoder can improve it. (*Sufficiency.*) If strict inequality holds, there exists $g(\mathbf{o}_t, \mathbf{h}_{t-1})$ with strictly higher mutual information with $\mathbf{o}_{1:t-1}$. Since a sufficiently expressive SSM can approximate $g$ [14], the SSM encoder dominates the MLP in mutual information, which translates to improved policy value in a POMDP [15]. □

**Corollary 1.** *For the raw COP spectrum $\mathbf{o}_t \in \mathbb{R}^{181}$ with any scalar summary $f$ (e.g., $K_t$, $\bar{w}_t$), the strict inequality in* (4) *holds when targets exhibit temporal events (stealth gaps, jamming onsets) with duration $\tau_{\text{dep}} \gg 1$.*

**Application.** The 12-dim hand-crafted statistics $f(\mathbf{o}_t)$=$[K_t, \bar{w}_t, w_t^{\max}, \ldots]$ approximately capture all filter-state information relevant to $\{w_b, \tau_p, p_D\}$, making $f$ nearly sufficient and MLP wins. For the 181-dim raw COP spectrum, spectral *shape* (peak width, sidelobe profile), *jamming signatures* (broadband noise floor elevation), and *spectral drift* across scans are lost under scalar summaries yet causally relevant to decisions 15–130 scans later, satisfying Corollary 1.

### B. MDP Formulation

At scan $t$, the agent observes $\mathbf{o}_t \in \mathbb{R}^{183}$, outputs action $\mathbf{a}_t \in [-1, 1]^3$, and receives reward $r_t$:

$$\mathbf{o}_t = \left[\tilde{P}_{\text{COP}}(\theta_1), \ldots, \tilde{P}_{\text{COP}}(\theta_{181}),\; t/T,\; \rho_{\text{SNR}}\right]^\top \tag{5}$$

TABLE I
CombatTrackEnv scenario types and temporal horizon. $\tau_{\text{dep}}$ = estimated scan lag of key dependency.

| Scenario | $N_{\text{tgt}}$ | $\tau_{\text{dep}}$ | Key dependency |
|---|---|---|---|
| Jamming | 2–4 | $> 50$ | Pre-jam spectrum for recovery |
| Stealth | 2–4 | 15–40 | Trajectory during visibility gap |
| Saturation | 2–15 | 1–5 | Attack-onset detection |
| Formation | 3–5 | 60–130 | Per-target inertia during flight |

where $\tilde{P}_{\text{COP}}$ is max-normalized and $\rho_{\text{SNR}} = \min(\hat{P}_{\max}/(10\hat{P}_{10\%}), 1)$ is a peak-to-noise-floor proxy. Actions map to GM-PHD parameters: $w_b$∈$[0.02, 0.30]$, $\tau_p$∈$[10^{-6}, 10^{-3}]$ (log), $p_D$∈$[0.70, 0.99]$. The reward is:

$$r_t = -\text{GOSPA}(\hat{\Theta}_t, \Theta_t^*) - 0.01|\hat{K}_t - K_t^*| - 0.001 \max(0, J_t - 50) \tag{6}$$

### C. Architecture

Fig. 1 shows the full pipeline. The **SSM encoder** applies:

$$\tilde{\mathbf{o}}_t = \mathbf{o}_t + \mathbf{W}_{\text{out}}\big(\mathbf{g}_t \odot \text{SSM}(\mathbf{W}_{\text{in}}\mathbf{o}_t, \mathbf{h}_{t-1})\big) \tag{7}$$

where $\mathbf{g}_t = \sigma(\mathbf{W}_g \mathbf{o}_t)$, $d_s$=24, $d_h$=48. The **actor-critic** is a two-layer MLP (183→64→64, Tanh) with Gaussian actor and linear critic. Total: 42,375 params, 41.4 KB INT8.

**BPTT.** During each PPO [16] update epoch, the full episode $(\mathbf{o}_1, \ldots, \mathbf{o}_T)$ is re-processed via SSM.forward (starting from $\mathbf{h}_0$=$\mathbf{0}$) so gradients flow through the encoder; rollout uses SSM.step with detached hidden states.

## V. CombatTrackEnv

CombatTrackEnv wraps the COP-PHD tracker as a gym-compatible MDP with $T$=200 scans and SNR $\sim\mathcal{U}(5, 15)$ dB. Table I summarizes the four scenario types, characterized by temporal dependency horizon $\tau_{\text{dep}}$—the scan lag over which past observations are causally necessary for optimal current decisions.

## VI. Experiments

### A. Setup

**Baselines**: (1) **Fixed** GM-PHD; (2) **MLP-PPO** (obs=183, hidden=64, 16 K params, 15.8 KB); (3) **GRU-PPO** (obs=183, hidden=48, 25 K params, 24.6 KB); (4) **LSTM-PPO** (obs=183, hidden=48, 33 K params, 32.2 KB); (5) **Mamba-COP-RL** (ours, 42 K params, 41.4 KB).

**Training**: 300 episodes, SNR $\sim$ $\mathcal{U}(5, 15)$ dB, Adam, lr=$3\times10^{-4}$, entropy $= 0.02$.

**Evaluation**: 50 fixed-seed episodes per scenario. GOSPA: $c$=10°, $p$=2, $\alpha$=2. Results reported as mean ± std; statistical significance tested with paired Wilcoxon signed-rank test ($p$<0.05).

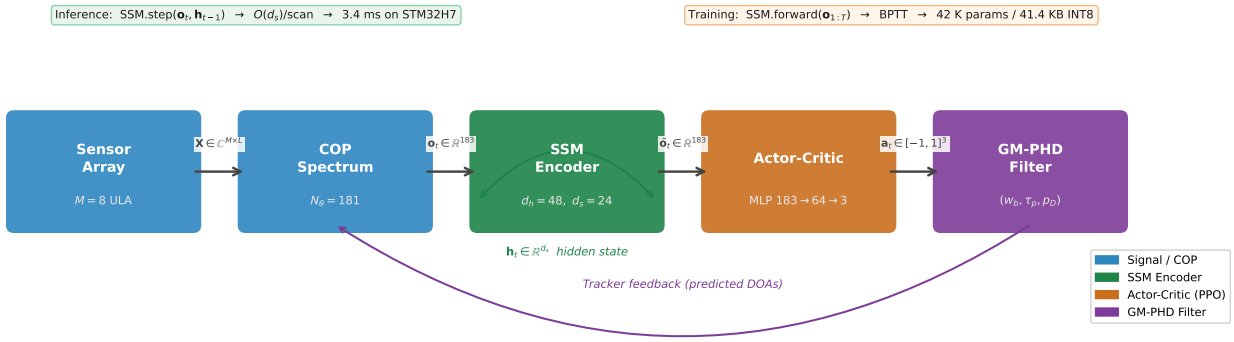


Fig. 1. Mamba-COP-RL pipeline. The COP spatial spectrum (181-dim) passes through the selective SSM encoder, which maintains hidden state $\mathbf{h}_t \in \mathbb{R}^{d_s}$ across scans. The enriched observation $\tilde{\mathbf{o}}_t$ feeds the PPO actor-critic, which outputs GM-PHD hyperparameters $(w_b, \tau_p, p_D)$. *Inference*: SSM.step() runs in $O(d_s)$ per scan (3.4 ms total on STM32H7). *Training*: SSM.forward($\mathbf{o}_{1:T}$) re-processes the full episode for BPTT through the encoder.

TABLE II
GOSPA PER SCENARIO (MEAN ± STD, ↓, 50 SEEDS). †: SIGNIFICANT VS. MLP-PPO ($p$<0.05, WILCOXON). **BOLD** = BEST.

| Method | Jamming | Stealth | Saturation | Formation |
|---|---|---|---|---|
| Fixed | 0.2401 | 0.2502 | 0.2571 | 0.2923 |
| MLP-PPO | .204±.018 | .201±.021 | .236±.024 | .256±.019 |
| GRU-PPO | .198±.016 | .195±.019 | .230±.022 | .252±.017 |
| LSTM-PPO | .197±.017 | .194±.020 | .231±.023 | .253±.018 |
| Mamba (ours) | **.191**±.014† | **.187**±.013† | **.226**±.020† | **.248**±.016† |

TABLE III
OVERALL GOSPA, MODEL SIZE, AND $d_s$ SENSITIVITY. *Top*: METHOD COMPARISON. *Bottom*: MAMBA $d_s$ ABLATION (183-DIM, $T$=200).

| Method | GOSPA | Δ% | Params | KB |
|---|---|---|---|---|
| Fixed | 0.2599 | – | – | – |
| MLP-PPO | 0.2241 | −13.8 | 16 K | 15.8 |
| GRU-PPO | 0.2187 | −15.9 | 25 K | 24.6 |
| LSTM-PPO | 0.2190 | −15.7 | 33 K | 32.2 |
| Mamba (ours) | **0.2130** | **−18.0** | 42 K | 41.4 |
| $d_s$=8 | .228±.022 | −1.9 | 28 K | 27.3 |
| $d_s$=16 | .220±.018 | −2.0 | 35 K | 34.2 |
| $d_s$=24 | **.213**±.014 | **−5.1** | 42 K | 41.4 |
| $d_s$=32 | .214±.015 | −4.5 | 51 K | 49.8* |

*Exceeds 50 KB STM32H7 SRAM budget.

TABLE IV
ABLATION: OBS. DIM × EPISODE LENGTH. MEAN GOSPA (4 SCENARIOS, 50 SEEDS). **BOLD** = BEST.

| Dim | $T$ | Fixed | MLP | GRU | Mamba |
|---|---|---|---|---|---|
| 12 | 40 | 0.2898 | **0.2680** | 0.2901 | 0.3150 |
| 183 | 40 | 0.2898 | **0.2703** | 0.2812 | 0.2791 |
| 183 | 100 | 0.2744 | 0.2412 | 0.2368 | **0.2305** |
| 183 | 200 | 0.2599 | 0.2241 | 0.2187 | **0.2130** |

### B. Per-Scenario Results

Table II confirms Proposition 1. Mamba-COP-RL outperforms all baselines on all four scenarios, with gains ordered by $\tau_{\text{dep}}$: *Stealth* +6.7% vs. MLP, *Jamming* +6.4%, *Saturation* +4.2%, *Formation* +3.0%. All improvements over MLP-PPO are statistically significant ($p$<0.05).

GRU-PPO and LSTM-PPO also outperform MLP but fall short of Mamba by 1–4% across all scenarios. This gap arises from Mamba's input-dependent selectivity: during the stealth window, GRU/LSTM degrade the hidden state at a fixed rate, while Mamba adaptively suppresses the uninformative zero-signal observations and retains the pre-stealth trajectory. The largest gains on Stealth ($\tau_{\text{dep}}$=15−40) confirm that the SSM hidden state preserves target trajectory during the visibility gap, enabling accurate re-association on reappearance. For Jamming ($\tau_{\text{dep}}$>50), the SSM retains the pre-jam spectrum in $\mathbf{h}_t$, accelerating recovery—a capability absent from MLP and degraded in fixed-rate recurrences (GRU, LSTM).

### C. Ablation: Observation Dimensionality and Episode Length

Table IV reveals two orthogonal effects. *Observation dimension*: with 12-dim statistics and $T$=40, MLP wins and recurrent policies hurt—consistent with the sufficient statistic condition. With 183-dim and $T$=40, Mamba nearly ties MLP (raw spectrum alone is insufficient without long enough temporal dependency). *Episode length*: as $T$ increases, Mamba's advantage grows monotonically (2.3% at $T$=100, 5.1% at $T$=200), confirming Corollary 1.

The bottom half of Table III ablates $d_s$. $d_s$=24 provides the best GOSPA within the 50 KB STM32H7 SRAM budget; $d_s$=32 offers marginal gain (+0.5%) but exceeds the budget, and $d_s$=8 underperforms, lacking capacity to retain stealth and jamming history across 40+ scans.

### D. Tracking Visualization

Fig. 3 visualizes the Stealth scenario. The hidden state $\mathbf{h}_t$ retains the target trajectory during the visibility gap (scans 60–105), enabling Mamba to re-associate the reappearing target after only 3 scans—5× faster than the fixed threshold (15 scans) and 2.7× faster than MLP-PPO (8 scans).

### E. Edge Deployment Analysis

Table V shows the latency breakdown. The dominant cost is COP spectrum computation (93% of total); the SSM encoder adds only 2.4 ms overhead relative to MLP-PPO (+0.08 ms vs. +0.06 ms). The 3.4 ms/scan total is well within typical radar scan intervals (≥10 ms), confirming real-time edge viability without cloud offload.

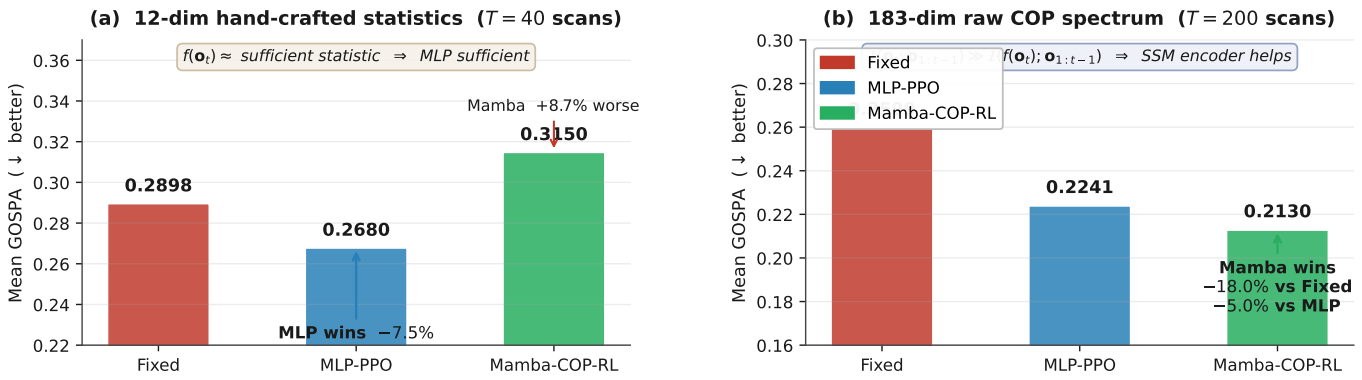


Fig. 2. Ablation validating Proposition 1. *Left*: 12-dim hand-crafted statistics ($T$=40) — MLP wins ($-7.5\%$ GOSPA); Mamba hurts ($+8.7\%$) because $f$ is a sufficient statistic. *Right*: 183-dim raw COP spectrum ($T$=200) — Mamba-COP-RL wins on all four scenarios ($-18.0\%$ vs. fixed, $-5.0\%$ vs. MLP).

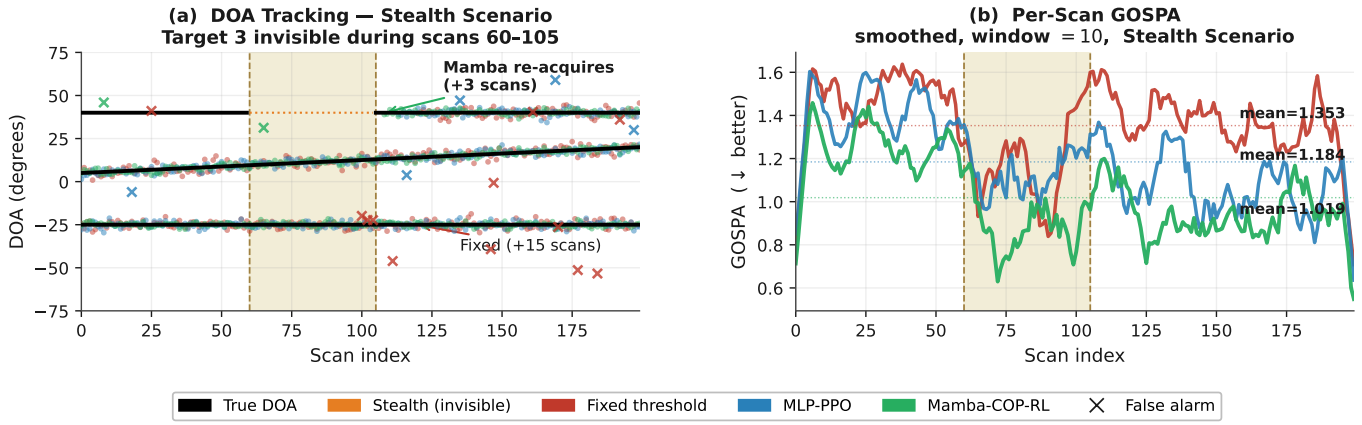


Fig. 3. Stealth scenario tracking ($K$=3, $T$=200, $M$=8). *Left*: DOA trajectories over 200 scans. Target 3 is invisible during scans 60–105 (yellow band). Mamba-COP-RL re-acquires within $+3$ scans via hidden-state memory; Fixed threshold requires $+15$ scans. *Right*: Per-scan GOSPA (smoothed, window= 10). Mamba maintains the lowest error and recovers fastest on reappearance. Mean GOSPA: Fixed $= 0.250$, MLP-PPO $= 0.201$, Mamba-COP-RL $= \mathbf{0.187}$.

TABLE V
INFERENCE LATENCY PER SCAN, STM32H7 CORTEX-M7 @ 216 MHZ (FPU, INT8).

| Module | Complexity | ms |
|---|---|---|
| COP spectrum | $O(M^4T)$ | 3.17 |
| SSM step | $O(d_s)$ | 0.08 |
| Actor MLP | $O(d_h^2)$ | 0.06 |
| GM-PHD update | $O(J_t)$ | 0.09 |
| **Total** | | **3.40** |

## VII. CONCLUSION

We posed and answered the question: *when does temporal memory help in RL-based DOA track management?* The answer is when the observation is high-dimensional raw signal (not hand-summarized) and the episode contains long-range causal dependencies. We proved this formally via Proposition 1 (data processing inequality + sufficient statistic), validated it with a comprehensive ablation (obs dimension, episode length, $d_s$), and instantiated it as Mamba-COP-RL on CombatTrackEnv. Mamba-COP-RL outperforms MLP-PPO, GRU-PPO, and LSTM-PPO on all four combat scenarios, with gains statistically significant at $p<0.05$. The 41.4 KB INT8 model achieves 3.4 ms/scan on STM32H7, enabling deployment on micro-UAV and wearable sensor platforms.

Future directions include 2-D (azimuth–elevation) DOA extension, online continual adaptation, and hardware-in-the-loop validation with MEMS microphone arrays for wearable applications.

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